

Model Question Paper I
INDUSTRIAL DRIVES & CONTROL

Time: 3 Hour

Max.Marks: 75

PART A

I. Answer **all** questions in one word or one sentence. Each question carries 1 mark.

1	Define an electric drive.	M1.01	R
2	Name an example for multi motor drive.	M1.02	R
3	List the methods of electric braking in DC motors.	M1.04	R
4	List any one application of AC drive.	M2.01	R
5	Enumerate two speed control methods of three phase induction motors.	M2.03	R
6	State the advantage of regenerative braking in the induction motor.	M2.04	R
7	Define the term continuous maximum rating of an electric motor.	M3.01	R
8	Name the motor suitable for low power (<1KW) pump drive applications.	M3.03	R
9	Define the term specific energy output.	M4.04	R

PART B

II. Answer any **eight** questions from the following, each question carries 3 marks.

1	Summarize any six factors affecting selection of electric drives.	M1.02	U
2	Explain various speed control methods of DC motors.	M1.03	U
3	Illustrate how rheostatic braking is achieved in DC series motors.	M1.04	U
4	Explain stator voltage control of single-phase induction motors using SCR.	M2.02	U
5	Explain any three advantages of synchronous motors.	M2.01	U
6	Illustrate dynamic braking method in single phase induction motor.	M2.04	U

7	An electric motor operates at full load of 100 KW for 10 minutes, at $\frac{3}{4}$ full load for next 10 minutes and at $\frac{1}{2}$ load for next 20 minutes, no load for the next 20 minutes and this cycle repeats continuously. Find the continuous rating of the suitable motor.	M3.01	A
8	Summarize three advantages and disadvantages of electric vehicles over internal combustion engine vehicles.	M3.03	U
9	Summarize any three requirements of an ideal traction system.	M4.01	U
10	A suburban train runs with an average speed of 36 km/h between two stations 2 km apart. Values of acceleration and retardation are 1.8 km/h/s and 3.6 km/h/s. Compute the maximum speed of the train assuming trapezoidal speed time curve.	M4.03	A

PART C

Answer **ALL** questions. Each question carries 7 marks.

III	Summarize the operation of single-phase half converter DC motor drive with necessary waveforms.	M1.03	U
	OR		
IV	Explain the operation of single-phase semi converter drives.	M1.03	U
V	Explain any three schemes for V/f control of three phase induction motor drives.	M2.03	U
	OR		
VI	Explain static rotor resistance control of three phase induction motors.	M2.03	U
VII	Summarize the application of electric drive in steel mills.	M3.02	U
	OR		
VIII	Explain electric drive system in textile mills.	M3.02	U
IX	Explain any six classes of motor duty.	M3.01	U
	OR		
X	Summarize any seven losses in the electric drive system.	M3.04	U
XI	An electric drive has an average speed of 42 km/h on a level track	M4.03	A

	between stations 1400 m apart. It is accelerated at 1.7 km/hr/sec and is braked at 3.3 km/hr/sec. Draw the speed time curve for the run.		
	OR		
XII	Develop an equation for traction effort for propulsion of the train.	M4.04	A
XIII	Illustrate the layout of an electric locomotive system.	M4.01	U
	OR		
XIV	Compare electric traction system using pure AC and DC series motors.	M4.01	U

Model Question Paper II
INDUSTRIAL DRIVES & CONTROL

Time: 3 Hour

Max.Marks: 75

PART A

I. Answer **all** questions in one word or one sentence. Each question carries 1 mark.

1	State the use of Sensing unit in an electric drive.	M1.01	R
2	Name an example for individual drive.	M1.02	R
3	List any two methods of braking in DC motors.	M1.04	R
4	List any one application of DC series motor drive.	M1.03	R
5	List any two breaking methods of three phase induction motor.	M2.04	R
6	List any two speed control methods of three phase induction motor.	M2.02	R
7	Enumerate two drives working under Continuous duty	M3.01	R
8	Name the motor suitable for high power (greater than 1KW) pump drive applications.	M3.03	R
9	Define the term Coefficient of Adhesion.	M4.01	R

PART B

IV. Answer any **eight** questions from the following, each question carries 3 marks.

1	Summarize any three advantages of Electric Drives.	M1.01	U
2	Explain Plugging method of DC motors.	M1.04	U
3	Compare Electric drives and mechanical Drives.	M1.01	U
4	Summarize any three features of AC Motor Drives.	M2.01	U
5	Explain the applications of synchronous motor drives.	M2.01	U
6	Illustrate Regenerative braking method in single phase induction motor.	M2.04	U
7	Explain any three losses in an Electrical Drive System.	M3.04	U
8	Illustrate block diagram of solar pump drives using induction motor drives.	M3.03	U
9	Summarize the factors affecting specific energy consumption.	M4.04	U
10	Explain tractive effort.	M4.01	U

PART C

Answer ALL questions. Each question carries 7 marks.

III	Explain the various components of Electric Drive with a block diagram.	M1.01	U
	OR		
IV	Explain the different factors for the choice of Electric Drives.	M1.01	U
V	Explain various schemes for speed control of three phase induction motor drives.	M2.03	U
	OR		
VI	Explain any two schemes for braking of three phase induction motor.	M2.04	U
VII	Summarize the application of electric drive in cement mills.	M3.02	U
	OR		
VIII	Explain electric drive system in sugar mills.	M3.02	U
IX	A Motor has to perform the following duty cycle 100 HP for 10 min No load for 5 min 60 HP for 8 min No load for 4 min which is repeated infinitely. Determine the suitable size of continuously rated motor. Also draw the curve for the load cycle.	M3.01	A
	OR		
X	A certain motor has to perform the following duty cycle: 100 KW for 10 minutes No load for 5 minutes 50 KW for 8 minutes No load for 4 minutes The duty cycle is repeated indefinitely. Draw the curve for the load cycle. Assuming that the heating is proportional to the square of the load, determine the suitable size of a continuously rated motor.	M3.01	A
XI	The distance between two stops is 1.2 km. A scheduled speed of 40 kmph is required to cover that distance. The stop is of 18-s duration. The values of the acceleration and retardation are 2 km/hr/sec and 3 km/hr/sec, respectively. Then, determine the maximum speed over the run. Assume a simplified trapezoidal speed–time curve.	M4.03	A
	OR		
XII	The distance between two stops is 5 km. A train has schedule speed of 50 kmph. The train accelerates at 2.5 km/hr/sec and retards 3.5 km/hr/sec and the duration of stop is 55 s. Determine the crest speed over the run assuming trapezoidal speed–time curve.	M4.03	A
XIII	Illustrate the different types of Passenger Services.	M4.02	U
	OR		
XIV	Explain different supply systems used in railway electrification.	M4.01	U